Project Proposal

Urban Concept Electric Vehicle

TEAM AVERERA

"Driven by innovation, Crafted with Precision"

This proposal outlines a potential collaboration with Team AVERERA, IIT (BHU) Varanasi, for fostering innovation in electric mobility through industry sponsors, mentorship, workshops, and engineering collaborations.

About Us

Team AVERERA is the energy-efficiency vehicle team of IIT (BHU) Varanasi participating in the Shell Eco-marathon Urban Concept category. We are dedicated to designing and developing high-efficiency electric vehicles. Our objective is not to build the fastest vehicle, but to answer a more important engineering question: *"How far can an electric vehicle travel using the least possible amount of energy?"*.

We have had the opportunity to represent IIT (BHU) Varanasi at multiple national/international events including the prestigious Shell Eco Marathon (Asia, Pacific). We have won multiple awards in the past including a podium finish in 2018 and 2019 in the EV Prototype Category; Carbon Footprint reduction awards in 2023 & 2025 and Technical Innovation Award in 2025.

Vision and Mission

Founded in 2013 and guided by the principle: "Leave the world better than you found it": Our vision is to develop future-ready EV engineers capable of designing efficient, reliable, and sustainable transportation technologies.

Our mission is to:

- Promote innovation in electric mobility.
- Establish laboratory-scale EV testing and characterization facilities
- Provide students with hands-on engineering experience, train students in industry-relevant electric vehicle engineering practices.
- Build strong collaborations between academia and industry.
- Contribute towards India's sustainable transportation ecosystem.
- Represent IIT (BHU) Varanasi at national and international engineering competitions such as the Shell Eco-marathon.

Project Goal

The goal is to design, fabricate and race an ultra-efficient EV, that would deliver over 230 km/kWh in the new urban concept car we are supposed to make. With this project, we aim to build the next generation of engineers who would push the barriers of EV. One major criticism EV industries today face is that the carbon footprint generated during extraction of batteries may be far more than those generated by the IC engine cars throughout their life. Hence EVs are not truly green. The only way to make EVs sustainable is to build an ultra-efficient EV by modifying the design or production technologies. Team AVERERA is dedicated to promoting sustainable mobility, and hence, a foundational step is building a single seater urban concept EV; and compete at an international scale.

Currently companies like Ather, Ola, Mahindra, Tata, BYD, Hyundai and Tesla are also working on improving the mileage and performance of EVs. In the two-wheeler industry the maximum efficiency is around 45 km/kWh and for cars it's around 13 km/kWh. Global student engineering teams—such as SZ Energy (Hungary), TUfast Eco Team (TU Munich) and Denmark Technological University (DTU) — have been working consistently for years and set international efficiency benchmarks at the Shell Eco-marathon. We are driven to surpass these international standards, bridging the gap between student innovation and commercial application to develop industry-relevant technologies that break the current limits of EV performance.

Why only Team AVERERA?

Team AVERERA holds a distinguished legacy at the Shell Eco-marathon, highlighted by a 2019 Global Championship in the Prototype EV category. Since transitioning to the Urban Concept class (2022), we have earned the Technical Innovation Award (2025) and dual Carbon Footprint Reduction Awards (2023, 2025). Driven by a rigorous daily R&D commitment, our multidisciplinary team is relentlessly engineering our next-generation vehicle.

Our work is guided by the deep domain expertise and industry networks of our faculty advisors: Dr. Amitesh Kumar (Mechanical Engineering), alongside Dr. Shyam Kamal and Dr. Sandip Ghosh (Electrical Engineering). Furthermore, IIT (BHU) actively champions our research by providing dedicated lab space, advanced manufacturing capabilities, and high-performance computing access for our complex simulations (to some extent). This strong institutional foundation is continually reinforced by ongoing technical mentorship from our engaged alumni network.

Contents

Executive Summary

Our current vehicle has demonstrated promising results yet. To push the boundaries further, Team AVERERA plans to establish dedicated research infrastructure and experimental facilities in five strategic domains:

1. Thermal Management
2. Battery Management Systems
3. High-Efficiency Powertrain Development
4. Suspension Optimization
5. Lightweight Composite Structures

These developments will lay the foundation for a permanent electric mobility research ecosystem within the institute, enabling continuous innovation in energy-efficient vehicle technologies while cultivating the next generation of engineers driving India's transition toward sustainable transportation

All the R&D related to the project spans across a period of 24 months and is divided into 3 phases.

1. Phase I: Basic understanding and numerical modelling of the problem
2. Phase II: Narrowing the field of focus to one among multiple options and optimization of the solution via simulations and laboratory trials.
3. Phase III: Final testing by integration into the vehicle.

Team AVERERA will adopt an iterative **"Design → Simulate → Prototype → Validate → Integrate"** development methodology to minimize technical and project risks.

This project, is not only a R&D project spanning 2 years, but goes beyond that developing a permanent research infrastructure at the institute which include:

- Thermal Characterization Testbench
- Battery Characterization & Cycling Bench
- Custom BMS Development Platform
- Motor Dynamometer Testbench
- Powertrain Validation Platform
- Experimental Suspension Test Rig
- Active/Passive Damping Research Platform
- Composite Material Characterization Facility
- Urban Concept EV Demonstrator Vehicle

Budget Summary

S. No.	Research Vertical / Activity	Consumables & Fabrication (₹)	Permanent Assets & Equipment (₹)	Total (₹)
1	Thermal Management Research	3,78,000	2,60,000	6,38,000
2	Battery Management Systems Research	5,32,000	3,10,000	8,42,000
3	Efficient Powertrain Development	5,46,400	8,12,000	13,58,400
4	Suspension Development & Testing	2,80,000	82,000	3,62,000
5	Lightweight Composite Development	2,56,000	6,10,000	8,56,000
6	Competition Expenses	29,50,000	__	29,50,000
	Grand Total	**49,42,400**	**20,74,000**	**70,16,400**

Companies can support by:

1. Financial sponsorship of full project/one or more R&D activity(es) or funding the competition expenses.
2. Sponsorship in kind by providing some parts/equipment or manufacturing the same for the team.
3. Technical mentorship: access to premium software used at the industry level for designing, simulations and numerical analysis; lectures/seminars/webinars; design reviews/evaluations.
4. Logistics support: Fund our delivery, import/export charges, team travel charges.

Project Breakdown

1. Thermal Management

The performance, efficiency, reliability, and safety of an electric vehicle are intrinsically linked to its thermal behaviour. Every major subsystem—including the battery pack, motor, motor controller, and power electronics—generates heat during operation. If not effectively managed, this heat can lead to performance degradation, accelerated component aging, reduced energy efficiency, and, in extreme cases, system failure. In an electric vehicle, every watt saved is additional distance travelled. Thermal management plays a crucial role in minimizing energy losses, preserving battery health, and ensuring that the vehicle operates at peak efficiency throughout its duty cycle. For ultra-energy-efficient vehicles such as Team AVERERA's Urban Concept EV, thermal management assumes even greater importance.

Why Thermal Management Matters

- Enhances battery efficiency and usable energy capacity.
- Improves battery lifespan by minimizing thermal degradation.
- Reduces electrical losses in motors and power electronics.
- Prevents thermal hotspots that may compromise safety.
- Improves overall vehicle reliability and consistency.
- Enables data-driven design decisions through thermal characterization and testing.
- Contributes directly to maximizing vehicle efficiency.

Recognizing these challenges, Team AVERERA aims to establish a dedicated thermal management research program focused on understanding heat generation mechanisms, characterizing thermal behaviour, validating cooling strategies, and developing data-driven thermal optimization methodologies for future vehicle platforms.

Prior Work/Experience

Our team has explored passive cooling systems in our previous iterations. Low-drag NACA submerged ducts were designed and integrated into the bodywork to route cooling airflow to the motor controller and battery pack without introducing a bluff-body inlet that would spike drag. Each duct was individually shaped and CFD-optimized for ramp angle, divergence angle, and depth ratio to maximize mass flow capture for a given drag penalty. We are yet to take it ahead for experimental verification. However, the CFD results indicate slight improvements in efficiency. This necessitates the need for coupling it with other passive/active cooling systems.

While these efforts established a foundational understanding of thermal behavior, a dedicated research framework involving thermal simulations, controlled testing, and cooling system optimization has not yet been implemented. The proposed project aims to bridge this gap and establish a structured thermal management research capability within Team AVERERA.

Plan of Action

Phase I: Thermal Characterization & Data Acquisition

Time: 2 months

Activities

Battery Thermal Characterization

- Measure cell temperatures under various discharge rates.
- Identify temperature gradients within the battery pack.
- Evaluate the relationship between temperature and battery efficiency.

Motor & Controller Thermal Characterization

- Monitor temperature rise during representative operating cycles.
- Identify peak thermal loads and hotspot locations.
- Determine steady-state and transient thermal behavior.

Instrumentation Development

- Installation of thermocouples and temperature sensors.

- Development of real-time data acquisition systems.
- Calibration and validation of thermal measurements.

Deliverables

- Thermal performance database.
- Heat generation estimates for major subsystems.
- Identification of critical thermal hotspots.
- Baseline thermal model for simulation studies.

Phase II: Cooling System Design & Optimization via CFD, FEA Simulations

Time: 12 months

Activities:

Thermal Simulation

- Development of thermal/mathematical models for:
  - Battery Pack
  - Motor
  - Motor Controller
- CFD analysis of airflow through cooling channels.

Cooling Architecture Development

- Design of passive/active cooling solutions.
- Design and optimization of cooling channel, cooling fins (or other modules, eg, Peltier).
- Investigation of heat sink configurations.
- Evaluation of airflow pathways within the vehicle body.

Prototype Development

- Fabrication of cooling system prototypes.
- Thermal interface material selection.
- Heat sink manufacturing and testing.

Deliverables

- Optimized cooling system design.
- CFD-validated cooling architecture.
- Prototype cooling components.
- Thermal simulation reports.

Phase III: Experimental Validation & Vehicle Integration

Time: 10 months

Activities

Thermal Test Bench Development

- Construction of a subsystem-level thermal testing setup.
- Controlled loading of battery and controller systems.
- Continuous thermal monitoring under simulated race conditions.

Thermal Imaging & Validation

- Infrared thermal imaging.
- Hotspot verification.
- Correlation between simulation and experimental results.

Vehicle-Level Testing

- Installation of cooling systems on the vehicle.
- Endurance testing.
- Thermal performance evaluation during full operating cycles.

Deliverables

- Validated thermal management system.
- Thermal performance reports.
- Vehicle integration documentation.
- Competition-ready cooling architecture.

Cost Breakdown

Item	Quantity	Unit Cost (₹)	Total Cost (₹)
Scale Model 3D Printing (Wind Tunnel)	5	25,000	1,25,000
NACA Duct Mold Fabrication (CNC)	4	27,000	1,08,000
Thermal Imaging Camera (FLIR)	1	90,000	90,000
Cooling Fins/Modules Development (Prototype Fabrication)	1	70,000	70,000
USB Data Acquisition System (Multi-channel)	1	70,000	70,000
Pitot-Static Tube & Differential Pressure Measurement System	1	30,000	30,000
Motor Thermal Testing Kit	1	40,000	40,000
Tuft Flow Visualization Kit	1	5,000	5,000
Smoke visualization setup	1	25,000	25,000
Custom testbench (insulated enclosure + heating elements + controllers + cooling system)	1	75,000	75,000
Total Consumables			**₹3,78,000**
Total permanent setup			**₹2,60,000**
Grand Total			**₹6,38,000**

2. Battery Management System

The battery pack is the single most critical subsystem in an electric vehicle, serving as the primary energy source and directly influencing vehicle range, efficiency, reliability, and safety. However, a battery pack is only as effective as the system managing it. A Battery Management System (BMS) continuously monitors, protects, and optimizes battery operation by tracking key parameters such as voltage, current, temperature, State of Charge (SOC), and State of Health (SOH).

For ultra-energy-efficient vehicles such as Team AVERERA's Urban Concept EV, maximizing the utilization of every watt-hour stored in the battery is essential. Even small inaccuracies in charge estimation, cell imbalance, or inefficient operating conditions can lead to a measurable reduction in vehicle range and competition performance. An advanced BMS not only safeguards the battery from unsafe operating conditions but also enables intelligent energy utilization, ensuring that the maximum possible energy is extracted from the pack without compromising battery longevity

Why Battery Management Matters

- Maximizes usable battery capacity and available energy.
- Improves vehicle range through accurate energy estimation and utilization
- Protects cells from overcharging, over-discharging, and excessive current draw.
- Maintains cell balancing to ensure uniform battery performance.
- Enhances battery lifespan by preventing accelerated degradation.
- Enables real-time monitoring and fault detection.
- Provides reliable State of Charge (SOC) and State of Health (SOH) estimation.
- Improves vehicle safety and operational reliability.
- Generates valuable battery performance data for future vehicle development.
- Contributes directly to maximizing vehicle efficiency and energy utilization.

Recognizing the central role of battery intelligence in vehicle performance, Team AVERERA aims to establish a dedicated Battery Management Systems research and experimentation facility.

Prior Work/Experience

While Team AVERERA has not previously undertaken dedicated Battery Management System research, the team has developed substantial expertise in vehicle electronics through the design, integration, and operation of motor controllers, power electronics, embedded systems, and vehicle instrumentation across multiple vehicle iterations.
Experience gained in PCB design, microcontroller programming, sensor integration, real-time data acquisition, and communication protocols has provided a strong technical foundation for advanced BMS development. These competencies position the team to systematically explore battery characterization, state estimation algorithms, cell balancing strategies, and intelligent battery monitoring systems.

Plan of Action

Phase I:Build a fully functional baseline Battery Management System (BMS)

The primary goals is to set the baseline by a building a BMS that safely monitors and protects the pack, utilizing basic passive balancing and Coulomb-counting State of Charge (SoC) estimation for the first year of competition.

- Monitoring: Implement voltage, current, and temperature sensing across the entire battery pack.
- Protection: Establish standard cut-offs for overvoltage, undervoltage, overcurrent, and overtemperature events.
- Cell Balancing: Utilize a passive (resistor-based) balancing method.
- State Estimation: Deploy basic Coulomb counting to estimate the SoC.
- Communication: Set up a fundamental CAN/UART data link to the vehicle controller.
- Dataset & Validation: Generate a charge/discharge cycling dataset on the bench to properly characterize the cells.

Phase II: BMS Development & Algorithm Design

Time: 12 months

Upgrade the baseline BMS to feature active cell balancing, model-based SoC/SoH (State of Health) estimation via Kalman filtering, and complete telemetry logging to target high efficiency scores.

- Monitoring: Upgrade to higher-resolution sampling and incorporate added diagnostics for early fault detection.
- Protection: Refine the safety threshold cut-offs and validate them against actual race-condition data.
- Cell Balancing: Transition to active balancing (capacitor or inductor-based) to recover more usable capacity.
- State Estimation: Implement model-based SoC/SoH estimation for improved accuracy when the system is under load.
- Communication: Introduce full telemetry logging and develop a live data dashboard for the driver and pit crew.
- Dataset & Validation: Collect on-track data across real race runs to validate the model estimates against real-world performance.

Deliverables

- Prototype custom BMS hardware.
- Embedded monitoring software.
- Protection and fault management framework.
- Real-time battery monitoring system.

Cost Breakdown

Item	Quantity	Unit Cost (₹)	Total Cost (₹)
Battery	4	25,000	1,00,000
Battery dataset generation rig (multi-channel cycler for cell characterisation data)	1	3,00,000	3,00,000
AFE (Analog Front-end) Evaluation Module (EVM)	1	55,000	55,000
STM32 development kits	10	2,000	20,000
PCB manufacturing (multiple iterations)	20	5,000	1,00,000
Protection circuitry & passive balancing hardware	1	35,000	35,000
Active balancing hardware	1	80,000	80,000
MCU platform + firmware development	_	30,000	30,000
Bench & vehicle integration testing	_	45,000	45,000
Telemetry & data-logging system	_	32,000	32,000
Miscellaneous electronics	_	45,000	45,000
Total Consumables			**₹3,10,000**
Total permanent setup			**₹5,32,000**
Grand Total			**₹8,42,000**

3. Efficient Powertrain Development

The powertrain serves as the energy conversion backbone of an electric vehicle, transforming electrical energy stored within the battery into mechanical energy that propels the vehicle. Its efficiency directly determines how effectively the available battery energy is utilized and, consequently, the distance that can be travelled per unit of energy consumed.

For ultra-energy-efficient vehicles such as Team AVERERA's Urban Concept EV, powertrain optimization extends beyond achieving adequate performance. Every watt lost due to motor inefficiencies, controller switching losses, drivetrain friction, or suboptimal operating conditions represents energy that could otherwise contribute to vehicle range. Even marginal improvements in motor and controller efficiency can result in significant gains in overall vehicle performance during endurance-based efficiency competitions.

Why Efficient Powertrain Development Matters

- Maximizes conversion of electrical energy into useful mechanical work.
- Reduces energy losses in motors, controllers, and drivetrain components.
- Improves overall vehicle efficiency and competition performance.
- Enables operation within peak motor efficiency regions.
- Improves controller performance through optimized switching strategies.
- Enhances vehicle reliability through component characterization and validation.
- Supports data-driven optimization of drivetrain architecture.
- Provides valuable performance data for future vehicle development.
- Reduces thermal losses and associated cooling requirements.
- Contributes directly to maximizing distance travelled per unit of energy consumed.

Recognizing the critical role of powertrain efficiency in achieving ultra-high vehicle efficiency, Team AVERERA aims to establish a dedicated powertrain research and experimentation facility focused on motor characterization, controller optimization, drivetrain efficiency mapping, and powertrain validation.

Prior Work/Experience

Team AVERERA has accumulated significant experience in electric powertrain integration through the development and operation of multiple vehicle iterations. The team has worked extensively with BLDC motor controllers, power distribution systems, vehicle telemetry, and real-time control architectures.

We have also worked extensively on genetic algorithms for optimizing transmission design (key parameter ,e.g, gear ratio) and driving strategy. It is based on modelling and simulations done in MATLAB and Simulink developed on track data obtained from GPS coordinates. Besides, extensive research has been done on components selection(sprag, chain sprocket, etc.) and DFM to ensure virtual and real-world outputs remain comparable.

Our work has not been able to isolate each factor and tune it specifically; thus, the optimum hasn’t been achieved. A structured research framework for powertrain characterization, efficiency mapping, controller optimization, and experimental validation has not yet been established. The proposed project seeks to bridge this gap by developing dedicated infrastructure and methodologies.

Plan of Action

<u>Phase I: Powertrain Characterization & Data Acquisition</u>

Time: 3 Months

Activities

<u>Motor Characterization</u>

- Current and voltage profiling under varying loads (from track data).
- Determination of optimal operating regions.
- Thermal performance characterization.
- Building a mathematical model to replicate the motor efficiency.

<u>Controller Characterization</u>

- Switching loss analysis.
- Efficiency evaluation under varying load conditions.
- Controller response and calibration studies.

Drivetrain Analysis

- Evaluation of transmission losses.
- Bearing and rolling resistance assessment.
- Chain/belt drive efficiency studies.
- Power flow analysis through the drivetrain.

Data Acquisition Infrastructure

- Development of motor testing instrumentation.
- Sensor integration and calibration.
- Real-time power monitoring setup.
- Automated data logging framework.

Deliverables

- Comprehensive motor performance database.
- Efficiency maps for motor and controller systems.
- Drivetrain loss characterization reports.
- Baseline powertrain models for optimization studies.

Phase II: Powertrain Optimization & Testbench Development

Time: 15 Months

Activities

Custom Motor design

- Design and optimization of custom motor
- Manufacturing

Powertrain Testbench Development

- Design and fabrication of motor dynamometer setup.
- Torque and speed measurement system integration.
- Power analyser integration.
- Controller testing platform development.

Controller Development & Optimization

- Controller calibration and tuning.
- Optimization of switching parameters.
- Energy-efficient drive strategies.
- Communication and telemetry integration.

System-Level Optimization

- Vehicle duty-cycle analysis.
- Powertrain operating strategy development.
- Energy consumption modelling.
- Optimization of gear ratios and drivetrain configuration.

Laboratory level HIL for vehicle

- Developing a system to isolate the effect of each subsystem on the vehicle performance.
- Testbench to study the effect of braking, turning losses, tyre wear etc.
- Fine tuning of all systems.

Deliverables

- Dedicated motor dynamometer testbench.
- Optimized motor and motor-controller configuration.
- Vehicle-specific powertrain operating strategy.
- Powertrain optimization framework.

Phase III: Validation & Vehicle Integration

Time: 6 Months
Activities
<u>Bench-Level Validation</u>

- Dynamometer testing under representative operating conditions.
- Efficiency validation of optimized configurations.
- Controller reliability assessment.
- Long-duration endurance testing.

<u>Vehicle Integration</u>

- Installation of optimized powertrain system.
- Telemetry integration for real-time monitoring.
- Calibration under actual operating conditions.

<u>Performance Validation</u>

- Vehicle energy consumption analysis.
- Competition-cycle simulation testing.
- Efficiency validation during track testing.
- Reliability and durability evaluation.

Deliverables

- Fully characterized and optimized powertrain system.
- Vehicle-integrated performance monitoring platform.
- Validated efficiency maps and operating strategies.
- Powertrain performance and reliability reports.

Cost Breakdown

Item	**Quantity**	**Unit Cost (₹)**	**Total Cost (₹)**
Custom gear set	3	11,000	33,000
Sprag clutch	6	7,000	42,000
Bearings	16	400	6,400
Front Transmission manufacturing	3	15,000	45,000
Rear Transmission manufacturing	3	60,000	1,80,000
Trailing arm suspension	3	14,000	42,000
Low Rolling Resistance Tyres	6	33,000	1,98,000
Custom BLDC/PMSM Motor	3	2,00,000	6,00,000
Motor controller manufacturing	2	10,000	20,000
Loading module	1	50,000	50,000
RPM Sensor	1	10,000	10,000
Voltage sensor	2	1,000	2,000
Torque sensor	1	70,000	70,000
Temperature transducer	6	5,000	30,000
T-slot cross iron rails	5	10,000	50,000
Total Consumables			**₹5,46,400**
Total permanent setup			**₹8,12,000**
Grand Total			**₹13,58,400**

4. Suspension Systems Development

The suspension system plays a critical role in determining the dynamic behaviour, stability, and energy efficiency of a vehicle. While conventional suspension systems are primarily designed to improve ride comfort and handling, ultra-energy-efficient vehicles demand a fundamentally different approach. In such vehicles, suspension design must balance ride quality, vehicle stability, rolling resistance, and energy losses associated with vibrations and tire-road interactions.

For Team AVERERA's Urban Concept EV, suspension development extends beyond traditional component selection. It presents an opportunity to investigate advanced vibration isolation technologies and novel suspension architectures capable of improving both vehicle dynamics and energy efficiency. By combining analytical modelling, simulation, experimental testing, and prototype development, the team aims to establish a research-driven framework for next-generation suspension systems.

Why Suspension System Development Matters

- Improves vehicle stability and handling characteristics.
- Reduces vibration-induced energy losses.
- Maintains optimal tire-road contact for reduced rolling resistance.
- Enhances driver comfort and vehicle controllability.
- Minimizes structural fatigue caused by repeated dynamic loading.
- Improves reliability of onboard electronics and battery systems.
- Enables optimization of vehicle dynamics for energy-efficient operation.
- Contributes directly to maximizing distance travelled per unit of energy consumed.

Recognizing the importance of vehicle dynamics in achieving ultra-high efficiency, Team AVERERA aims to establish a dedicated Suspension Systems Research and Experimentation Facility focused on vibration isolation, dynamic stiffness tuning, advanced damping technologies, and intelligent suspension control methodologies.

Prior Work / Experience

Team AVERERA has previously designed, manufactured, and integrated suspension systems for multiple vehicle iterations, gaining practical experience in suspension geometry design, load estimation, component selection, structural analysis, and vehicle testing. The team has developed expertise in CAD modelling, finite element analysis, chassis integration, and vehicle dynamics simulations.

Additionally, the team has undertaken preliminary investigations into vibration isolation concepts and compliant structural systems through academic and research activities. Exposure to advanced concepts such as dynamic stiffness modulation, quasi-zero stiffness mechanisms, and passive vibration control technologies has created a strong foundation for structured suspension research.

While previous efforts have focused on achieving reliable vehicle operation, a dedicated research framework involving suspension characterization, vibration testing, dynamic stiffness optimization, and advanced control methodologies has not yet been established. The proposed project aims to address this gap and create an in-house capability for next-generation suspension development.

Plan of Action

Phase I: Suspension Characterization & Dynamic Modelling

Time: 6 Months

Activities

Vehicle Dynamics Modelling

- Development of dynamic models.
- Ride and handling simulations under representative operating conditions.
- Identification of dominant vibration modes.
- Frequency response analysis of existing suspension systems.

CAD Design and FEA simulations

- Choice of suspension system (trailing arm/ wishbone/Mac-pherson strut) and accordingly deciding suspension parameters
- CAD design and structural analysis

Suspension Characterization

- Spring stiffness characterization.
- Damping coefficient estimation.
- Tire stiffness evaluation.
- Ride frequency and transmissibility analysis.

Deliverables

- Vehicle dynamic models.
- Frequency response and transmissibility reports.
- Baseline suspension design.

Phase II: Suspension Development, Testbench Creation, Experiments

Time: 15 Months

Activities

Dynamic Stiffness Tuning

- Investigation of variable stiffness suspension architectures.
- Development of tunable spring-damper systems.
- Optimization of suspension natural frequencies.
- Vehicle-specific ride and stability tuning.

Quasi-Zero Stiffness (QZS) Suspension Research

- Analytical modelling of QZS mechanisms.
- Design and simulation of compliant QZS systems.
- Fabrication of prototype QZS suspension assemblies.
- Experimental validation of vibration isolation performance.

Advanced Passive Suspension Technologies

- Investigation of eddy current damping systems.
- Development of non-contact passive damping concepts.
- Comparative evaluation against conventional dampers.
- Integration studies for lightweight vehicle applications.

Active & Semi-Active Control Systems

- Development of active suspension control architecture.
- Sensor integration and control strategy development.
- Real-time suspension response optimization.
- Evaluation of energy requirements versus performance benefits.

Suspension Testbench Development and Testing

- Design and fabrication of quarter-car suspension test rig.
- Actuator-based road excitation system.
- Dynamic load simulation capability.
- Instrumentation and data acquisition integration

Laboratory Validation

- Frequency sweep testing.
- Vibration transmissibility measurements.
- Dynamic stiffness validation.
- Comparative evaluation of passive, semi-active, and active systems.

Deliverables

- Suspension dynamics testbench.
- Tunable suspension prototypes.
- QZS suspension demonstrator.
- Eddy current damping evaluation platform.

- Active/semi-active suspension control framework.
- Static and dynamic analysis and final architecture selection

<u>Phase III: Validation & Vehicle Integration</u>
Time: 3 Months
Activities
<u>Manufacturing</u>

- Transit from lab scale prototypes to actual suspension

<u>Vehicle Integration</u>

- Installation of optimized suspension architecture.
- Sensor and telemetry integration.
- Vehicle dynamics calibration.

<u>Track Validation</u>

- Ride and handling evaluation.
- Endurance testing under representative operating conditions.
- Energy loss assessment due to vehicle vibrations.
- Stability and reliability validation.

<u>Performance Evaluation</u>

- Comparison against baseline suspension systems.
- Efficiency impact assessment.
- Driver feedback and operational evaluation.

Deliverables

- Fully validated suspension architecture.
- Vehicle-integrated suspension monitoring platform.
- Dynamic performance database.
- Suspension optimization reports.
- Competition-ready suspension system.

Cost Breakdown

Item	Qty	Unit Cost (₹)	Total Cost (₹)
Experimental Vibration Test Rig Fabrication	1	50,000	50,000
Accelerometers	4	8,000	32,000
Precision Springs (Various Stiffnesses)	10	2,500	25,000
QZS Mechanism Prototype Fabrication (iterations)	4	5,000	20,000
Flexure-Based Suspension Prototype Fabrication (iterations)	4	15,000	60,000
Passive Damping Prototype Materials (Eddy Current & Other Concepts)	__	40,000	40,000
Active Damping Electronics (MCU, Sensors, Drivers)	__	65,000	65,000
Full-Scale Vehicle Suspension Manufacturing (Final Design Only)	1	75,000	75,000
Misceallaneous (Bearings, Fasteners, Fixtures & Consumables)	__	15,000	15,000
Total Consumables			**₹2,80,000**
Total permanent setup			**₹82,000**
Grand Total			**₹3,62,000**

5. Lightweight Composite Structures Development

As vehicle efficiency is directly linked to energy consumption, reducing body weight while maintaining sufficient stiffness and durability is a key engineering challenge. Composite materials offer a unique opportunity to tailor mechanical properties to specific design requirements. Unlike conventional metallic structures, composites allow engineers to strategically place material only where it is needed, enabling significant weight reduction without compromising performance. However, achieving an optimal composite design requires a careful balance between weight, stiffness, manufacturability, cost, impact resistance, and long-term durability.

Why Composite Development Matters

- Reduces overall vehicle mass and energy consumption.
- Improves strength-to-weight and stiffness-to-weight ratios.
- Enhances structural efficiency through tailored fiber orientations.
- Reduces inertia, improving vehicle responsiveness and efficiency.
- Provides opportunities for material and manufacturing innovation.
- Generates valuable material characterization data for future vehicle development.
- Contributes directly to maximizing vehicle efficiency and range.

Recognizing the importance of lightweight structures in achieving ultra-high efficiency, Team AVERERA aims to establish a dedicated Composite Materials Research and Manufacturing Facility focused on material characterization, structural optimization, advanced layup methodologies, and lightweight body construction.

The Energy & Manufacturing Bottleneck

Traditional aerospace-grade composites rely on autoclave curing, which demands massive thermal and electrical energy for prolonged temperature and pressure cycles. Furthermore, conventional wet-layup or prepreg processes require complex vacuum bagging (involving peel plies, breathers, and sealant tapes) that is highly labor-intensive, generates significant consumable waste, and is heavily dependent on operator skill.

The Composite Welding Challenge

Conventional thermoset matrices (like standard epoxies) form irreversible, cross-linked chemical networks; they cannot be melted or re-formed. Joining these panels traditionally forces a reliance on mechanical fasteners—which introduce localized stress concentrations and parasitic weight—or secondary structural adhesives, which require meticulous surface preparation, long cure times, and are susceptible to environmental degradation. Thermoplastic matrices solve this, as they can be fusion-bonded, but the engineering challenge lies in delivering highly localized heat to the interface without degrading the surrounding fiber-matrix architecture or causing polymer degradation.

Prior Work / Experience

Team AVERERA has previously designed and manufactured CFRP outerbody and other composite body panels and lightweight structural components for its competition vehicles.
The team has also developed expertise in CAD modelling and structural analysis of laminates.

While these efforts have enabled successful vehicle construction; systematic research into composite material characterization and experimental work laminate optimization, sandwich structures, and advanced lightweight manufacturing techniques has not yet been undertaken. The proposed project aims to establish a structured research framework that will support future composite development activities.

Plan of Action

Phase I: Research Infrastructure Development & Process Characterization

Time: 6 Months

Activities

Testbench Setup

- Commission lab infrastructure.

- Install localized heat-presses.
- Setup OOA consolidation.
- Calibrate ultrasonic generators.

Material Screening

- Carbon fiber laminates.
- Fiberglass laminates.
- Carbon-glass hybrid composites.
- Carbon-basalt composites.
- Carbon-Kevlar composites.
- Off the shelf composites (UHMWPE, Aerogel-based lightweight composite, Natural fiber composites)
- Core materials including foam and honeycomb structures.

Mechanical Characterization

- Tensile testing.
- Compression testing.
- Flexural testing.
- Impact resistance evaluation.
- Interlaminar failure studies.
- Fatigue behaviour assessment.
- Stiffness-to-weight comparison.

Thermal Profiling

- Map melt temperatures (T_m).
- Determine glass transitions (T_g).
- Conduct DSC analysis.
- Test thermal degradation.

Baseline Auditing

- Process baseline laminates.
- Audit baseline energy.
- Record cycle times.
- Measure electrical draw.

Tooling & Handling

- Prototype lightweight molds.
- Test high-density foams.
- Establish handling SOPs.
- Define pre-consolidation steps.

Deliverables

- Operational research testbenches.
- Thermal parameter database.
- Baseline energy metrics.
- Comparative performance reports.
- Manufacturing process guidelines.
- Lightweight tooling prototypes.

Phase II: OOA Optimization & Manufacturability

Time: 9 Months

Activities

OOA(Out of Autoclave) Process Development

- Optimize OOA cycles.
- Identify minimum dwell.
- Develop matched-die thermoforming.
- Eliminate bagging consumables.

Smart Tooling Integration

- Design localized susceptors.
- Run computational analyses.
- Target specific heating.
- Reduce thermal mass.

Core Structure Consolidation

- Bond thermoplastic skins.
- Integrate foam cores.
- Integrate honeycomb structures.
- Execute rapid-heating cycles.

Prototype Development

- Structural panel prototypes.
- Hybrid laminate demonstrators.
- Sandwich panel demonstrators.
- Lightweight structural concepts.

Deliverables

- OOA manufacturing guidelines.
- Consumable-free forming jigs.
- Process efficiency maps.
- Optimized sandwich panels.
- Reduced cycle times.

Phase III: Welding & Structural Integration

Time: 8 Months

Activities

Localized Heat Delivery

- Engineer Energy Directors.
- Optimize ultrasonic frequencies.
- Evaluate induction flux.
- Test embedded susceptors.

Joint Analysis Strategy

- Simulate Heat-Affected Zones.
- Conduct lap-shear testing.
- Eliminate mechanical fasteners.
- Remove structural adhesives.

Prototype Body Assembly

- Execute localized welding.
- Assemble body panels.
- Integrate vehicle chassis.
- Weight verification.
- Structural stiffness assessment.
- Surface quality evaluation.
- Vehicle integration testing.

Deliverables

- Welded technology demonstrator.
- Ultrasonic welding SOPs.
- Induction welding SOPs.
- Mass reduction report.
- Assembly time validation.
- Full-scale research-derived aeroshell.
- Vehicle-integrated composite body.
- Composite manufacturing documentation.

Cost Breakdown

Item	Qty	Unit Cost (₹)	Total Cost (₹)
Thermal Consolidation & OOA Processing Setup	1	65,000	65,000
Composite Fusion Welding Infrastructure	1	85,000	85,000
Fibers and fabrics	$5m^2$	15,200	76,000
Aerogel	1kg	20,000	20,000
Thermoplastic Matrices (e.g., PA6, PEEK films)	10kg	3,000	30,000
Thermal Profiling & Baseline Testing Materials	1	15,000	15,000
Small-Scale Mold Fabrication & Tooling	1	15,000	15,000
Epoxy Resins, Hardeners & Consumables	10kg	2,500	25,000
Foam & Honeycomb Core Materials	1 roll	15,000	15,000
Coupon Fabrication & Prototype Materials	1 set	20,000	20,000
Smart Susceptors & OOA Consolidation Jigs	1 set	25,000	25,000
Energy Directors & Embedded Susceptors	1 set	10,000	10,000
Vacuum Bagging Kit	1 set	15,000	15,000
Structural Validation & Prototype Iterations	__	20,000	20,000
Rims	6	15,000	90,000
Outer Body	1	3,20,000	3,20,000
Miscellaneous Consumables & Workshop Expenses	-	20,000	20,000
Total consumables			**2,56,000**
Total permanent setup			**6,10,000**
Grand Total			**8,66,000**

6. Competition Expenses – Shell Eco Marathon

To represent IIT-BHU on an international scale, the vehicle shall be taken to Shell Eco Marathon, an international competition with 70+ participating countries. The 2027 edition is scheduled to be held in Qatar. The venue changes every 3 years. The goal of the competition is to build the most efficient vehicle. We shall participate in the Urban Concept EV category. The competition would give an international exposure to the team, enhancing their experience. With the developed vehicle the team would participate in at least 2 years

Cost Breakdown

Subsystem	Item/Service	Qty.	Unit Cost (₹)	Total Cost (₹)
Logistics	Vehicle Shipping	2 yrs	10,00,000	20,00,000.
Travel	Team travel fare	10 x 2yrs	45,000	9,00,000
Miscellaneous	__	__	50,000	50,000
			Grand Total	**₹29,50,000**

Long Term Impact

- Establishment of a dedicated electric vehicle research ecosystem at IIT (BHU) focused on energy-efficient mobility technologies.
- Development of permanent research infrastructure, including test benches, characterization facilities, simulation frameworks, and experimental datasets that can be utilized by future student teams and researchers.
- Creation of indigenous expertise in critical EV domains such as thermal management, battery management systems, powertrain optimization, suspension engineering, and lightweight composite structures.
- Enhanced hands-on learning opportunities for students through exposure to the complete engineering development cycle, from concept generation and simulation to manufacturing, testing, and validation.
- Contribution to the development of industry-ready engineers equipped with practical experience in emerging electric vehicle technologies.
- Promotion of interdisciplinary collaboration across mechanical, electronics, electrical, material science and chemical engineering.
- Generation of reusable engineering methodologies, design practices, and technical documentation that will accelerate future vehicle development efforts.
- Strengthening of industry-academia collaboration through research-driven development and technology validation activities.
- Advancement of research in energy-efficient transportation systems, lightweight structures, intelligent energy management, and sustainable engineering practices.
- Improvement of institutional capabilities in electric mobility research, creating opportunities for future projects, collaborations, publications, and funding initiatives.
- Contribution towards India's transition to sustainable mobility by promoting technologies that maximize energy utilization and reduce transportation energy consumption.
- Creation of a long-term platform for innovation, research, and skill development that will continue delivering value well beyond the completion of the current Urban Concept Electric Vehicle project.

Risk Assessment & Mitigation

Team AVERERA will adopt an iterative **"Design → Simulate → Prototype → Validate → Integrate"** development methodology to minimize technical and project risks. Laboratory-scale experimentation will be used extensively before committing resources to full-scale manufacturing, ensuring that only validated concepts are incorporated into the final vehicle. Regular design reviews, milestone tracking, and documentation practices will further ensure that project objectives are achieved within the planned timeline and budget. Yet, there remain chances of failure (as in any other research).

Risk Category	Potential Risk	Mitigation Strategy
Technical Development	Experimental concepts such as QZS suspensions, advanced BMS algorithms, or novel composite architectures may not achieve the expected performance.	Multiple concepts will be evaluated in parallel. Proven baseline designs will be retained as fallback options for vehicle integration.
Research Uncertainty	Simulation results may not correlate perfectly with experimental outcomes.	Experimental validation will be conducted at each stage to verify models and refine design assumptions before vehicle implementation.
Manufacturing	Defects during composite fabrication, machining, or assembly may require redesign or rework.	Prototype iterations and intermediate validation checks will be performed before full-scale manufacturing.
Procurement & Supply Chain	Delays in procurement of sensors, electronics, composite materials, or custom-manufactured parts.	Procurement activities will begin early in the project timeline, with multiple vendor options identified for critical components.
Budget Constraints	Increase in material or equipment costs due to market fluctuations.	Budget estimates include contingency provisions and preference will be given to in-house manufacturing and institute facilities wherever feasible.
Equipment Availability	Limited availability of testing facilities, workshop resources, or laboratory equipment.	Testing schedules will be planned in advance and alternative institute facilities will be identified for critical activities.
Vehicle Integration	Individual subsystems may perform well independently but face compatibility challenges during vehicle integration.	Incremental subsystem integration and validation will be conducted throughout development rather than waiting until final assembly.
Data Acquisition & Testing	Sensor failures, inaccurate measurements, or incomplete datasets may affect research outcomes.	Calibration procedures, redundant measurements, and periodic verification of instrumentation will be implemented.
Timeline Management	Delays in research, fabrication, or testing activities may impact project completion. Further, examination schedules, internships or placements may hinder the performance of team individuals.	Project milestones will be monitored regularly, with buffer periods incorporated into the development schedule.
Team Continuity	Student graduation and turnover may result in knowledge loss during the project lifecycle.	Comprehensive documentation, design repositories, and knowledge transfer sessions will be maintained throughout the project.

The project primarily involves moderate technical and execution risks typical of research and development activities. However, these risks are mitigated through phased development, extensive prototype validation, utilization of existing institute infrastructure, and the availability of proven fallback solutions.

Types of Partners

To achieve our engineering and competitive objectives, Team AVERERA seeks to establish mutually beneficial partnerships across the following tiers:

Partner Category	Expected Contribution / Services Rendered
Financial Partners	Direct monetary grants or funding to subsidise research, vehicle manufacturing, and team operational expenditures.
Manufacturing & Fabrication Partners	Specialized machining, rapid prototyping (e.g., 3D printing), welding, and assembly services for custom vehicle components.
Material & Component Providers	Provision of essential raw materials (e.g., composites, aerospace-grade alloys) and specialized parts or equipments (e.g., motors, batteries, microcontrollers).
Technology & Software Partners	Licensing for advanced engineering software (CAD/CAM/CAE), data acquisition systems, and performance simulation tools.
Logistics & Supply Chain Partners	Freight, shipping, travel and handling services to safely transport the vehicle, tools, and equipment to domestic or international competition venues.
Knowledge & Testing Partners	Access to industry expertise, technical mentorship, and professional testing facilities (e.g., track access, dynamometers, wind tunnels).

Deliverables to Partners

Team AVERERA views this as a collaborative partnership that enables innovation in sustainable mobility. It is committed to:

Meaningful Technical Collaboration

- Opportunity to engage with students working on advanced EV technologies.
- Access to non-confidential project outcomes, performance data, and technical findings.
- Access to dataset generated during the testing of various equipments and/or findings made on custom testbenches.
- Opportunity to propose industry-relevant engineering challenges for student exploration.

Brand Visibility & Recognition

- Company logo placement on the Urban Concept Electric Vehicle.
- Logo placement on team apparel, pit setup, and promotional materials.
- Dedicated acknowledgment on Team AVERERA's website and social media platforms.
- Recognition during technical exhibitions, demonstrations, and institute events.

Talent & Recruitment Access

- Direct interaction with highly motivated students possessing hands-on experience in electric vehicle development.
- Opportunities for internships, mentorship programs, and recruitment engagement.
- Early access to a multidisciplinary talent pool with expertise in design, simulation, manufacturing, electronics, controls, and testing.

Institutional & Outreach Benefits

- Recognition as a contributor to sustainable mobility and student innovation.
- Visibility across IIT (BHU)'s academic and student community.
- Association with a nationally and internationally recognized engineering competition.

Besides, upon completion of the project, sponsors may be provided with:

- Project completion and utilization reports.
- High-level technical summaries of research activities.
- Vehicle performance highlights and competition outcomes.
- Documentation of infrastructure and facilities developed through the project.
- Impact assessment demonstrating the utilization of sponsored resources.

Previous Partners

Team AVERERA has actively collaborated with many companies/startups in the past. Team AVERERA boasts of its alumni in multiple distinguished companies, which we actively keep leveraging for collaboration. Over the past decade, we have had the fortune to partner with companies like Tata Motors, GoPro, HP. Nike, MATLAB, drobot and many more.

Conclusion

This project goes beyond mere competitions. The requested support will directly enable the creation of research infrastructure, experimental facilities, prototype development programs, and the final competition vehicle. More importantly, it will establish a long-term foundation for innovation in electric mobility, energy-efficient transportation, and sustainable engineering at IIT (BHU) Varanasi. By supporting Team AVERERA, partners will not only contribute to the development of a world-class energy-efficient vehicle but also help cultivate the next generation of engineers working toward India's clean mobility future.